

Development Plan

Project: AI Study Material Generator

Team Name : CSE4104-7C-T03

The development of the AI Study Material Generator will follow a phased approach to ensure systematic implementation, testing, and deployment of all project features.

Development Roadmap

Week	Activities	Expected Outcome
Week 05	Project setup, repository configuration, finalize UI/UX design	Project environment ready
Week 06	Backend development (Authentication APIs, User Management APIs)	User authentication system completed
Week 07	Frontend development (Landing Page, Login, Registration, Dashboard UI)	Basic user interface completed
Week 08	Database implementation and integration with backend	Database connected successfully
Week 09	Document upload module and file management implementation	Users can upload and manage documents
Week 10	AI integration using OpenAI/Gemini APIs	AI services integrated successfully
Week 11	Implementation of Summary, MCQ, Flashcard, and Quiz generation features	Core AI features completed
Week 12	AI Chatbot implementation and history management	Chatbot feature completed
Week 13	Testing, debugging, and performance optimization	Stable and optimized system
Week 14	Final deployment, documentation, and presentation preparation	Project ready for final submission

Module Development Sequence

Phase 1: Project Initialization

- Create GitHub repository
- Setup frontend and backend environments
- Configure database connection
- Prepare project structure

Phase 2: Backend Development

- Authentication Module
- User Management Module
- File Upload Module
- Document Management Module

Phase 3: Frontend Development

- Landing Page
- Login Page
- Registration Page
- Dashboard
- Profile Page
- AI Interaction Pages

Phase 4: Database Development

- Design collections/tables
- Implement CRUD operations
- Configure MongoDB Atlas

Phase 5: AI Integration

- Integrate OpenAI/Gemini APIs
- Implement document processing
- Generate summaries
- Generate MCQs
- Generate flashcards
- Generate quizzes
- Implement AI chatbot

Phase 6: Testing and Deployment

- Unit Testing
 - Integration Testing
 - Bug Fixing
 - Performance Testing
 - Deployment on cloud platform
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Expected Final Outcome

By the end of the semester, the team aims to deliver a fully functional AI-powered web application that enables students to:

- Upload study documents
- Generate AI-based summaries
- Generate MCQs and quizzes
- Create flashcards automatically
- Interact with an AI chatbot
- Download generated study materials

The system will provide an efficient and intelligent learning experience for students.